

PEACEABLE QUEENS ON EVEN TORI: AN EXACT PARITY FORMULA

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ABSTRACT. Let $t(n)$ be the largest m for which m black and m white queens can be placed on the $n \times n$ toroidal chessboard with no black–white attack. We prove that, for every positive integer q ,

$$t(2q) = \max_{(r_0, r_1, c_0, c_1) \in \{0, \dots, q\}^4} \min\{r_0 c_1 + r_1 c_0, (q - r_0)(q - c_0) + (q - r_1)(q - c_1)\}.$$

The maximizing construction separates the two square parities and extends the plaid idea of Clinch, Drescher, Huynh and Saffidine. The upper bound combines an exact moment relaxation, rational branch certificates, a hand-checkable algebraic finish for $q \geq 130$, and a rerunnable exact integer cut-envelope computation for the finite range. In particular,

$$\lim_{\substack{n \rightarrow \infty \\ 2|n}} \frac{t(n)}{n^2} = \frac{2 - \sqrt{3}}{2} = 0.1339745 \dots$$

We also prove the odd values $t(15) = 20$ and $t(17) = 28$ by exhaustive finite searches with separately written audits.

1. INTRODUCTION

A peaceable placement consists of equally many black and white queens, with no queen attacking a queen of the other colour; attacks within an army are allowed. The problem goes back to Ainley [1]. On the ordinary board its asymptotic density remains open [3]; on the toroidal board the rows, columns and diagonals wrap around, and the problem becomes translation invariant. Its maxima form OEIS sequence A279405 [4].

Clinch, Drescher, Huynh and Saffidine [3, 2] introduced *plaid* constructions on even tori. They obtained asymptotic lower density 0.1339 and upper density 0.1402, while for odd orders they proved the upper bound $t(n) \leq n^2/8$. Their local search also found, on even tori, placements that recolour a few rows and columns of an optimal plaid (their *swaps*) and exceed the best plaid by at most two queens in every case they examined; writing $P(n)$ for the largest value of $\min(|B|, |W|)$ over all (a, b) -plaids, they conjectured that $t(n) \leq P(n) + 2$ for every even n [2, Conjecture 5.1]. The following theorem determines the even problem exactly; Remark 11 relates it to $P(n)$ and to that conjecture.

Definition 1. For a positive integer q , put

$$(1) \quad H(2q) := \max_{(r_0, r_1, c_0, c_1) \in \{0, \dots, q\}^4} \min\{r_0 c_1 + r_1 c_0, (q - r_0)(q - c_0) + (q - r_1)(q - c_1)\}.$$

Theorem 2. For every positive integer q ,

$$t(2q) = H(2q).$$

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Corollary 3. *Let $\kappa = 4 - 2\sqrt{3}$ and $\gamma = 1 - 1/\sqrt{3}$. For every positive integer q ,*

$$\kappa q^2 - \gamma q - \frac{1}{4} \leq H(2q) \leq \kappa q^2.$$

Consequently, for even n ,

$$t(n) = \frac{2-\sqrt{3}}{2} n^2 + O(n), \quad \lim_{\substack{n \rightarrow \infty \\ 2|n}} \frac{t(n)}{n^2} = \frac{2-\sqrt{3}}{2} = 0.1339745\dots,$$

and $t(n) \leq \lfloor (2 - \sqrt{3})n^2/2 \rfloor$.

Corollary 4. *Every even-indexed term $t(2q)$ of A279405 is computable from (1); for example,*

$$t(16) = 32, \quad t(18) = 42, \quad t(20) = 51, \quad t(40) = 210.$$

Theorem 5. $t(15) = 20$ and $t(17) = 28$.

Theorem 2 implies that there always exists an optimal *parity-separated* line colouring: the diagonal colours reserve one square parity for each army, leaving the four row and column parity counts in (1). This family contains the classical plaids but is strictly larger; for example it gives $H(20) = 51$, whereas direct optimization of the classical two-parameter subfamily gives 50.

For the upper bound we first replace queens by a colouring of the four line families. On an even torus, conditioning on row and column parities yields a 42-equation moment relaxation. Two exact rational branch certificates localize every possible counterexample to a small neighbourhood of the parity-separated profile. Two support inequalities then reduce that neighbourhood to a short algebraic argument for $q \geq 130$. The remaining orders are covered by a rerunnable exact integer cut-envelope computation, apart from five small orders settled by direct exhaustive search. Section 7 gives a proof-complete account of the searches for Theorem 5. Appendix A describes the released certificates, audits and reproduction commands. All proof decisions use integer or rational arithmetic.

2. THE FORMULA AND THE LOWER BOUND

2.1. Line colourings. Throughout, $n \geq 2$, all coordinates lie in $\mathbb{Z}/n\mathbb{Z} = \{0, 1, \dots, n-1\}$, and the board is $G = (\mathbb{Z}/n\mathbb{Z})^2$. Through a cell (r, c) pass four *lines*: row r , column c , diagonal $d = r - c$ and antidiagonal $a = r + c$, each family indexed by $\mathbb{Z}/n\mathbb{Z}$. Two queens attack each other exactly when they share a line. For $R, C, D, A \subseteq \mathbb{Z}/n\mathbb{Z}$ put

$$B(R, C, D, A) = \{(r, c) \in G : r \in R, c \in C, r - c \in D, r + c \in A\},$$

and $W(R, C, D, A) = B(R^c, C^c, D^c, A^c)$, complements taken family by family.

Lemma 6 (Line-colouring equivalence). *Let $m \geq 1$. A peaceable placement of m black and m white queens on the $n \times n$ torus exists if and only if there are $R, C, D, A \subseteq \mathbb{Z}/n\mathbb{Z}$ with $|B| \geq m$ and $|W| \geq m$.*

Proof. ($\Rightarrow$) Colour every line carrying a black queen black and every other line white; no line carries queens of both colours, else those two queens attack. Every black queen then lies on four black lines and every white queen on four white lines, and the queens of one colour occupy distinct cells.

($\Leftarrow$) Place black queens on any m cells of B and white queens on any m cells of W . The two sets are disjoint, and a black–white attack would need a line lying simultaneously in a black set and in its complement. $\square$

Consequently $t(n) = \max_{R,C,D,A} \min(|B|, |W|)$, and the queens have disappeared. Complementing all four sets exchanges B and W, so $\min(|B|, |W|)$ is invariant under complementation; we use this freely.

Lemma 7 (Incidence). *Let ℓ, ℓ' be lines of two distinct families. Unless they form a diagonal-antidiagonal pair, they meet in exactly one cell. A diagonal d and an antidiagonal a meet in the fibre*

$$\varphi^{-1}(d, a), \quad \varphi(r, c) = (r - c, r + c).$$

Every point in this fibre satisfies $2r = a + d$ and $2c = a - d$; these doubled congruences are equivalent to the original pair only when n is odd. The fibre has one cell for odd n . For even n it has two cells, differing by $(n/2, n/2)$, when $d \equiv a \pmod{2}$, and is empty otherwise.

Proof. The first statement is immediate (a row r and a column c meet in (r, c) ; a row r and a diagonal d in $(r, r - d)$; and so on). For the second, consider $\varphi(r, c) = (r - c, r + c)$ on G . Its kernel is $\{(r, c) : r = c, 2r = 0\}$, trivial for odd n and of order 2 for even n , generated by $(\frac{n}{2}, \frac{n}{2})$. For even n every nonempty fibre thus has two elements and the image has $n^2/2$ elements; every image point satisfies $d \equiv a \pmod{2}$, and there are exactly $n^2/2$ such pairs, so the image is precisely that set. $\square$

Lemma 8 (Budget). *If $|B| \geq m$ and $|W| \geq m$ for some (R, C, D, A) , then the same holds for a configuration satisfying*

$$|R| + |C| + |D| + |A| \leq 2n.$$

Proof. Complementing all four sets exchanges B and W and replaces the sum of their sizes by $4n$ minus that sum. $\square$

For two line families X, Y , let $I(X, Y)$ denote the number of cells whose lines in both families are black.

Lemma 9 (Triple identity). *Choose three line families with black sets of sizes x, y, z . If U and U' count the cells whose three lines are all black and all white, respectively, then*

$$(2) \quad U + U' = n^2 - n(x + y + z) + I(X, Y) + I(X, Z) + I(Y, Z).$$

In particular, $|B|, |W| \geq m$ implies that the right-hand side is at least $2m$.

Proof. For the three black-line indicators u, v, w on the cells,

$$uvw + (1 - u)(1 - v)(1 - w) = 1 - u - v - w + uv + uw + vw.$$

Summing gives (2); moreover, $B \subseteq U$ and $W \subseteq U'$. $\square$

2.2. The parity-separated construction. Now let $n = 2q$. The diagonal and antidiagonal labels through (r, c) have the same parity, namely that of $r + c$. Colouring both diagonal families black on one label parity and white on the other therefore reserves one square parity for each army. Only the row and column choices remain.

Split each of the four families by label parity: for $i \in \{0, 1\}$ let $r_i = |\{\rho \in R : \rho \equiv i\}|$, and likewise c_i, d_i, a_i ; each parity class of $\mathbb{Z}/2q\mathbb{Z}$ has exactly q labels, so all eight counts lie in $\{0, \dots, q\}$.

Proposition 10 (Parity separation). *Let $q \geq 1$ and $0 \leq r_0, r_1, c_0, c_1 \leq q$. Choose $R \subseteq \mathbb{Z}/2q\mathbb{Z}$ consisting of any r_0 even and any r_1 odd labels, C of any c_0 even and c_1 odd labels, and set $D = A = \{\text{odd labels}\}$. Then*

$$|B| = r_0 c_1 + r_1 c_0, \quad |W| = (q - r_0)(q - c_0) + (q - r_1)(q - c_1).$$

Consequently $t(2q) \geq H(2q)$.

Proof. A cell (r, c) is black-homogeneous iff $r \in R$, $c \in C$ and both $r - c$ and $r + c$ are odd, i.e. iff $r \in R$, $c \in C$ and r, c have opposite parities; there are $r_0c_1 + r_1c_0$ such cells. It is white-homogeneous iff $r \notin R$, $c \notin C$ and $r + c$ is even, i.e. r, c of equal parity; there are $(q - r_0)(q - c_0) + (q - r_1)(q - c_1)$ such cells. Now apply Lemma 6 with $m = \min(|B|, |W|)$ and maximize over the four counts. $\square$

Remark 11. The construction extends the plaid mechanism of [2, §3]. Up to exchanging parity classes, the classical two-parameter subfamily has profiles (u, q, v, q) and hence capacity

$$P(2q) = \max_{u, v \in \{0, \dots, q\}} \min\{q(u + v), (q - u)(q - v)\}.$$

The four counts in Proposition 10 are independent, so its family is larger. Maximizing the two-variable expression gives $P(20) = 50$ and $P(24) = 72$, whereas $H(20) = 51$ (profile $(7, 2, 1, 7)$) and $H(24) = 74$ (profile $(8, 2, 1, 9)$). The swapped plaids of [2, §5] recolour rows and columns of a plaid; the example in [2, Figure 6] reaches 74 on the 24-torus, which is $H(24)$. A direct computation gives $H(2q) - P(2q) \in \{0, 1, 2\}$ for all $q \leq 200$: the difference is 0 at 146 of these orders, 1 at 33 (first at $n = 20$) and 2 at 21 (first at $n = 24$). Theorem 2 therefore confirms Conjecture 5.1 of [2] for every even $n \leq 400$ and replaces the inequality by an equality. We have not proved $H(2q) \leq P(2q) + 2$ for every q . At $n = 16$, the profile $(0, 6, 6, 0)$ gives 36 black-homogeneous and 32 white-homogeneous cells.

n	2	4	6	8	10	12	14	16	18	20
$H(n)$	0	2	4	8	12	18	25	32	42	51
n	22	24	26	28	30	32	34	36	38	40
$H(n)$	64	74	90	101	120	133	153	170	190	210

TABLE 1. Exact even values through order 40. The entries through order 14 agree with A279405, accessed July 27, 2026.

2.3. Values and size of H .

Lemma 12 (Continuous upper bound). *Let $q > 0$ be real and let $r_0, r_1, c_0, c_1 \in [0, q]$ be real. Put*

$$X = r_0c_1 + r_1c_0, \quad Y = (q - r_0)(q - c_0) + (q - r_1)(q - c_1).$$

Then $\min(X, Y) \leq \kappa q^2$, where $\kappa = 4 - 2\sqrt{3}$. In particular $H(2q) \leq \lfloor \kappa q^2 \rfloor = \lfloor \frac{2-\sqrt{3}}{2}(2q)^2 \rfloor$.

Proof. Normalize: $\tilde{r}_i = r_i/q$, $\tilde{c}_i = c_i/q$, and set

$$\rho = \tilde{r}_0 + \tilde{r}_1, \quad \chi = \tilde{c}_0 + \tilde{c}_1, \quad \xi = 1 - \rho, \quad \eta = 1 - \chi, \quad z = (\tilde{r}_0 - \tilde{r}_1)(\tilde{c}_0 - \tilde{c}_1).$$

Expanding,

$$(3) \quad \frac{X - Y}{q^2} = \rho + \chi - 2 - z, \quad \frac{X + Y}{q^2} = 2 - \rho - \chi + \rho\chi = 1 + \xi\eta,$$

and $|z| \leq \rho\chi$ because $|\tilde{r}_0 - \tilde{r}_1| \leq \tilde{r}_0 + \tilde{r}_1 = \rho$ and likewise for the columns.

If $\xi\eta \leq 0$ then $\min(X, Y) \leq \frac{X+Y}{2} \leq \frac{q^2}{2} < \kappa q^2$.

The map $(r_0, r_1, c_0, c_1) \mapsto (q - r_0, q - r_1, q - c_1, q - c_0)$ preserves the domain, exchanges X and Y , and replaces (ξ, η) by $(-\xi, -\eta)$. So we may assume $\xi, \eta \geq 0$.

Suppose first $X \geq Y$. Then $z \leq \rho + \chi - 2$ by (3), and $z \geq -\rho\chi$, so $2 - \rho - \chi \leq \rho\chi$, i.e. $\xi + \eta \leq 1 - \xi - \eta + \xi\eta$, i.e.

$$(4) \quad 2\xi + 2\eta - \xi\eta \leq 1.$$

Put $w = \sqrt{\xi\eta} \in [0, 1]$. By AM-GM, $4w - w^2 \leq 2\xi + 2\eta - \xi\eta \leq 1$, so $w^2 - 4w + 1 \geq 0$ and hence $w \leq 2 - \sqrt{3}$. By (3),

$$\min(X, Y) \leq \frac{X + Y}{2} = q^2 \frac{1 + w^2}{2} \leq q^2 \frac{1 + (2 - \sqrt{3})^2}{2} = (4 - 2\sqrt{3})q^2.$$

Now suppose $Y \geq X$, so $\min(X, Y) = X$, and assume for contradiction $X > \kappa q^2$. Then $X + Y \geq 2X > 2\kappa q^2$, so by (3) $\xi\eta > 2\kappa - 1 = (2 - \sqrt{3})^2$ and $w > 2 - \sqrt{3}$. On the other hand $X/q^2 = \tilde{r}_0\tilde{c}_1 + \tilde{r}_1\tilde{c}_0 \leq (\tilde{r}_0 + \tilde{r}_1)(\tilde{c}_0 + \tilde{c}_1) = \rho\chi$, because the two omitted products $\tilde{r}_0\tilde{c}_0$ and $\tilde{r}_1\tilde{c}_1$ are nonnegative, and $\rho\chi = (1 - \xi)(1 - \eta) \leq (1 - w)^2$ by AM-GM. Therefore

$$\kappa < \frac{X}{q^2} \leq (1 - w)^2 < (1 - (2 - \sqrt{3}))^2 = (\sqrt{3} - 1)^2 = 4 - 2\sqrt{3} = \kappa,$$

a contradiction. Finally, $H(2q)$ is an integer maximum of $\min(X, Y)$ over a subset of the domain, so $H(2q) \leq \lfloor \kappa q^2 \rfloor$. $\square$

Lemma 13. *Let $s = \sqrt{3} - 1$, so $\kappa = s^2 = 4 - 2\sqrt{3}$, and let $\gamma = 1 - 1/\sqrt{3} = s/(1 + s)$. Then for every $q \geq 1$,*

$$H(2q) \geq \kappa q^2 - \gamma q - \frac{1}{4}.$$

Proof. For an integer $0 \leq P \leq 2q$ take the profile $(r_0, r_1, c_0, c_1) = (0, \lfloor P/2 \rfloor, \lceil P/2 \rceil, 0)$. Then

$$X = \left\lfloor \frac{P}{2} \right\rfloor \left\lceil \frac{P}{2} \right\rceil = \left\lfloor \frac{P^2}{4} \right\rfloor, \quad Y = q \left(q - \left\lceil \frac{P}{2} \right\rceil \right) + \left(q - \left\lfloor \frac{P}{2} \right\rfloor \right) q = 2q^2 - qP,$$

so $H(2q) \geq \min(\lfloor P^2/4 \rfloor, 2q^2 - qP)$. The two branches cross where $P^2/4 = 2q^2 - qP$, i.e. at $P = 2sq$, with common value $s^2 q^2 = \kappa q^2$.

Write $2sq = m + \theta$ with $m = \lfloor 2sq \rfloor$ and $\theta \in [0, 1)$. Taking $P = m \leq 2sq$ gives $X \geq P^2/4 - \frac{1}{4} = \kappa q^2 - sq\theta + \theta^2/4 - \frac{1}{4} \geq \kappa q^2 - sq\theta - \frac{1}{4}$ and $Y = 2q^2 - qm = \kappa q^2 + q\theta \geq \kappa q^2$, so the value is at least $\kappa q^2 - sq\theta - \frac{1}{4}$. Taking $P = m + 1 \geq 2sq$ gives $X \geq P^2/4 - \frac{1}{4} \geq \kappa q^2 - \frac{1}{4}$ and $Y = \kappa q^2 - q(1 - \theta)$, so the value is at least $\kappa q^2 - q(1 - \theta) - \frac{1}{4}$. Hence

$$H(2q) \geq \kappa q^2 - q \min(s\theta, 1 - \theta) - \frac{1}{4} \geq \kappa q^2 - q \max_{0 \leq \theta < 1} \min(s\theta, 1 - \theta) - \frac{1}{4},$$

and the inner maximum is attained at $s\theta = 1 - \theta$, with value $s/(1 + s) = 1 - 1/\sqrt{3} = \gamma$. $\square$

Lemmas 12 and 13 give Corollary 3 once Theorem 2 is proved. Numerically $\gamma = 0.42264\dots$, so $H(2q)$ deviates from κq^2 by less than $\frac{1}{2}q + \frac{1}{4}$.

3. THE UPPER BOUND: RELAXATION AND CANONICAL CHAMBER

From here to the end of Section 5, $n = 2q$ and we prove $t(2q) \leq H(2q)$ for $q \geq 130$. Fix a configuration (R, C, D, A) with black count $B = |B|$ and white count $W = |W|$, and let

$$(r_0, r_1, c_0, c_1, d_0, d_1, a_0, a_1) \in \{0, \dots, q\}^8$$

be its eight parity-split outer counts, with normalization $x = (r_0, \dots, a_1)/q \in [0, 1]^8$.

3.1. The 42-equation moment relaxation. Partition the board into four *row/column parity blocks* $G_{ij} = \{(r, c) : r \equiv i, c \equiv j \pmod{2}\}$, each of q^2 cells. Inside G_{ij} every cell has square parity $e = i \oplus j$, so its diagonal and antidiagonal labels both have parity e ; and each of the four line families restricted to the relevant parity class has exactly q labels.

For each block introduce 16 nonnegative variables

$$p_{ij}(\varepsilon_R, \varepsilon_C, \varepsilon_D, \varepsilon_A), \quad (\varepsilon_R, \varepsilon_C, \varepsilon_D, \varepsilon_A) \in \{0, 1\}^4,$$

intended as q^{-2} times the number of cells of G_{ij} whose row, column, diagonal and antidiagonal are black (1) or white (0) as indicated. Write $E_{ij}[\cdot]$ for the corresponding expectation. A genuine configuration satisfies the following 42 equations, in which $e = i \oplus j$.

Lemma 14 (Soundness of the moment system). *For every configuration and every block (i, j) ,*

$$(5) \quad \sum p_{ij} = 1; \quad E_{ij}[\varepsilon_R] = \frac{r_i}{q}, \quad E_{ij}[\varepsilon_C] = \frac{c_j}{q}, \quad E_{ij}[\varepsilon_D] = \frac{d_e}{q}, \quad E_{ij}[\varepsilon_A] = \frac{a_e}{q};$$

$$(6) \quad E_{ij}[\varepsilon_R \varepsilon_C] = \frac{r_i c_j}{q^2}, \quad E_{ij}[\varepsilon_R \varepsilon_D] = \frac{r_i d_e}{q^2}, \quad E_{ij}[\varepsilon_R \varepsilon_A] = \frac{r_i a_e}{q^2},$$

$$E_{ij}[\varepsilon_C \varepsilon_D] = \frac{c_j d_e}{q^2}, \quad E_{ij}[\varepsilon_C \varepsilon_A] = \frac{c_j a_e}{q^2};$$

and, for each square parity $e \in \{0, 1\}$, the aggregate equation

$$(7) \quad \sum_{i \oplus j = e} E_{ij}[\varepsilon_D \varepsilon_A] = \frac{2 d_e a_e}{q^2}.$$

These are $4 \cdot 10 + 2 = 42$ equations, whose right-hand sides involve exactly 22 distinct products of outer counts. Moreover

$$(8) \quad \frac{B}{q^2} = \sum_{i,j} p_{ij}(1, 1, 1, 1), \quad \frac{W}{q^2} = \sum_{i,j} p_{ij}(0, 0, 0, 0).$$

Proof. The normalization and the four marginals in (5) hold because within G_{ij} each of the four families' relevant parity classes is swept uniformly: for instance the map $G_{ij} \rightarrow (\mathbb{Z}_q)$, $(r, c) \mapsto$ (index of r within its parity class), has all fibres of size q .

For (6) we need each pair of families other than $\{D, A\}$ to fibre uniformly over G_{ij} . Writing $r = 2r' + i$, $c = 2c' + j$ with $r', c' \in \mathbb{Z}_q$, the pair (row, column) corresponds to (r', c') , the pair (row, diagonal) to $(r', r' - c')$, (row, antidiagonal) to $(r', r' + c')$, (column, diagonal) to $(c', r' - c')$ and (column, antidiagonal) to $(c', r' + c')$. Each of these is, up to a block-dependent translation, an invertible linear map $\mathbb{Z}_q^2 \rightarrow \mathbb{Z}_q^2$ of determinant ± 1 , hence a bijection; so all q^2 label pairs occur exactly once and the corresponding moment factorizes into the product of the two black-line densities.

The pair $\{D, A\}$ does not fibre uniformly within a block: $(r', c') \mapsto (r' - c', r' + c')$ has determinant 2. But by Lemma 7 each pair (d, a) of labels of equal parity e meets in exactly two cells, and these two cells (r, c) , $(r + q, c + q)$ lie in the two blocks with $i \oplus j = e$ (in the same block iff q is even, in different blocks otherwise — in either case in the union). Summing over the two blocks therefore counts each equal-parity label pair exactly twice, giving (7). Finally (8) is the definition. $\square$

Remark 15. The failure of the $\{D, A\}$ pair to factorize *per block* is not cosmetic: for $4 \mid n$ the two lifts of an equal-parity (d, a) pair land in the *same* block, so per-block independence of the diagonal families is false and 2-adic information survives parity conditioning. Replacing (7) by four per-block equations would be unsound. This is the reason the relaxation has 42 and not 44 rows.

Definition 16. For $x \in [0, 1]^8$ let $\text{OPT}_{42}(x)$ be the maximum of t subject to: $p \geq 0$; the 42 equations of Lemma 14 with the right-hand sides evaluated at x (that is, r_i/q replaced by x -coordinates and each product $r_i c_j / q^2$ by the corresponding product of x -coordinates); and $t \leq \sum_{ij} p_{ij}(1, 1, 1, 1)$, $t \leq \sum_{ij} p_{ij}(0, 0, 0, 0)$.

The maximum is well defined. For each block and each x , four independent Bernoulli indicators with the prescribed marginals satisfy all block moments; summing the two relevant blocks also gives each aggregate $\{D, A\}$ equation. Thus the p -feasible set is nonempty. It is compact, and after eliminating t the objective is the minimum of two continuous linear functions of p , so the maximum is attained.

Lemma 14 immediately gives the key inequality

$$(9) \quad \frac{\min(B, W)}{q^2} \leq \text{OPT}_{42}(x)$$

for every configuration with normalized outer profile x .

3.2. The canonical chamber.

Lemma 17 (Chamber). *Every configuration can be transformed, without changing $\{B, W\}$ or the multiset of homogeneous cell counts, into one whose normalized profile x satisfies*

$$(10) \quad r_0 \leq r_1, \quad c_0 \leq c_1, \quad r_0 + r_1 \leq c_0 + c_1, \quad d_0 + d_1 \leq a_0 + a_1, \quad \sum_{i=1}^8 x_i \leq 4.$$

Proof. Complementing all four line sets exchanges B and W and replaces $\sum x_i$ by $8 - \sum x_i$, so the budget can be enforced. Translating the board by one unit in the row direction, $(r, c) \mapsto (r+1, c)$, exchanges the two row parity classes (and simultaneously the two diagonal and the two antidiagonal classes); translating by one unit in the column direction does the same for the columns. Composing the two if necessary, we may sort $r_0 \leq r_1$ and $c_0 \leq c_1$. Transposing, $(r, c) \mapsto (c, r)$, exchanges rows with columns and fixes the antidiagonals while negating the diagonals, hence orders the row and column totals. Finally $(r, c) \mapsto (r, -c)$ fixes the row parities, permutes the column labels within their parity classes, and exchanges the diagonal with the antidiagonal family, so it orders $d_0 + d_1 \leq a_0 + a_1$. Each of these is an affine bijection of the board mapping lines to lines, so B and W are preserved (or exchanged, in the complementation step). $\square$

Each step preserves what the earlier steps achieved. Complementation comes first, and the remaining maps permute labels within each family, so the budget row is kept. The transpose exchanges the row counts with the column counts and therefore maps the pair of conditions $r_0 \leq r_1$, $c_0 \leq c_1$ to itself; the final reflection fixes all four row and column parity counts and so preserves the first three ordering inequalities. We therefore assume (10) from now on.

4. THE TWO BRANCH CERTIFICATES AND THE ESCAPE COMPARISON

4.1. What the certificates are. For a box $[l, u] \subseteq [0, 1]^8$, replace each of the 22 products $x_i x_j$ occurring in the right-hand sides of the moment equations by a fresh variable y_{ij} constrained by the four McCormick envelope inequalities

$$(11) \quad \begin{aligned} -y_{ij} + l_i x_j + l_j x_i &\leq l_i l_j, & -y_{ij} + u_i x_j + u_j x_i &\leq u_i u_j, \\ y_{ij} - u_i x_j - l_j x_i &\leq -u_i l_j, & y_{ij} - l_i x_j - u_j x_i &\leq -l_i u_j, \end{aligned}$$

which are exactly the expansions of $(x_i - l_i)(x_j - l_j) \geq 0$, $(u_i - x_i)(u_j - x_j) \geq 0$, $(u_i - x_i)(x_j - l_j) \geq 0$ and $(x_i - l_i)(u_j - x_j) \geq 0$. We also retain the redundant valid bounds

$$l_i l_j \leq y_{ij} \leq u_i u_j;$$

these are used when maximizing the partially dualized Lagrangian. Together with the budget row, the four chamber rows of (10), and the two rows $t \leq B/q^2$, $t \leq W/q^2$, this gives a linear programme in the $8 + 22 + 64 + 1 = 95$ variables (x, y, p, t) with 95 non-bound inequality rows and 42 equality rows, whose optimum dominates $\text{OPT}_{42}(x)$ for every x in the box.

Lemma 18 (Dual-leaf soundness). *Consider the linear programme above on a box: maximize t over $z = (x, y, p, t)$ subject to the inequality rows $Az \leq b$ (McCormick, budget, chamber and the two objective rows), the 42 moment equalities $Ez = e$, the box bounds on (x, y) , and $p \geq 0$. Let $\mu \leq 0$ and ν be rational multipliers for the inequality and the equality rows, and form the partial Lagrangian*

$$L(z) = t + \mu^\top (Az - b) + \nu^\top (Ez - e).$$

Suppose that, after collecting terms, the coefficient of t in L vanishes and the coefficient vector ω of the atom variables satisfies $\omega \leq 0$, so that $L(z) = \ell_0 + \ell(x, y) + \omega^\top p$ with $\ell_0 = -\mu^\top b - \nu^\top e$ and ℓ linear. (Here b, e are the constraint right-hand sides, local to this lemma.) Then every feasible z satisfies

$$t \leq \ell_0 + \max_{(x, y) \in \text{box}} \ell(x, y),$$

and the maximum is computed exactly by taking each coordinate at the box endpoint matching the sign of its coefficient in ℓ .

Proof. If z is feasible then $\mu^\top (Az - b) \geq 0$, because componentwise both factors are nonpositive, and $\nu^\top (Ez - e) = 0$; hence $t \leq L(z)$. Since $\omega \leq 0$ and $p \geq 0$, we get $L(z) \leq \ell_0 + \ell(x, y)$. Finally, a linear function on a product of intervals is maximized coordinate by coordinate at interval endpoints. $\square$

A *branch certificate* is a rooted binary tree. A split node carries a coordinate index and an exact rational cut, and its two children are the parent box intersected with $x_d \leq c$ and with $x_d \geq c$ (the children overlap on the cutting hyperplane, which is harmless and gives an exhaustive cover). A leaf is one of three kinds:

- a *McCormick-dual leaf*, carrying an exact rational multiplier vector for the inequality rows (nonpositive) and an unrestricted one for the equality rows, together with the resulting exact rational bound u . The verifier reconstructs the partial Lagrangian, checks that the reduced cost of t vanishes and that the reduced costs of all 64 atom variables are ≤ 0 (so that the inner maximum over $p \geq 0$ is finite), evaluates the outer and product variables at their better box endpoints, recomputes the bound itself, and checks that it equals the stored u and satisfies the required threshold — that is, it verifies precisely the hypotheses of Lemma 18, whose conclusion certifies u as an upper bound for the relaxation on the leaf box;
- an *exact empty leaf*, carrying one of five reasons (**rsort**, **csort**, **rcsum**, **dasum**, **budget**). Each is a purely interval-arithmetic obstruction to the chamber inequalities: e.g. **rsort** asserts $\max(l_{r_1}, l_{r_0}) > u_{r_1}$, which contradicts $r_0 \leq r_1$; and **budget** asserts that the sum of the four group minima forced by the preceding steps exceeds 4. These are exact rational emptiness tests, never solver infeasibility claims;

- a *plaid leaf*, asserting that the box lies wholly inside one of the four rational plaid neighbourhoods. With $s = \sqrt{3} - 1$, the four templates are obtained by choosing a square parity and a row parity; a box qualifies when, coordinate by coordinate, the template-0 coordinates have upper endpoint $\leq \frac{1}{10}$, the template-1 coordinates have lower endpoint $\geq \frac{19}{20}$, and the two template- s coordinates lie in $[\frac{3}{5}, \frac{7}{8}]$.

Lemma 19 (Branch-certificate inference). *Suppose every terminal box of a branch certificate is either empty, contained in a declared exceptional region, or carries a valid dual upper bound u . Then the root box is the union of the exceptional terminal boxes and boxes on which the relaxation has value at most the corresponding u .*

Proof. Induct upward from the leaves. At a split $x_d \leq c$ or $x_d \geq c$, the two closed child boxes cover the parent. The stated alternatives are therefore preserved at every internal node and hence at the root. $\square$

Definition 20. The canonical rational plaid neighbourhood is

$$(12) \quad N : \quad 0 \leq r_0, c_0, d_1, a_1 \leq \frac{1}{10}, \quad \frac{3}{5} \leq r_1, c_1 \leq \frac{7}{8}, \quad \frac{19}{20} \leq d_0, a_0 \leq 1$$

(in normalized coordinates). After the chamber restrictions (10), it is the only one of the four templates that survives.

4.2. The two certificates.

Theorem 21 (Global localization). *For every $x \in [0, 1]^8$ satisfying the chamber inequalities (10) and lying outside N ,*

$$\text{OPT}_{42}(x) \leq \frac{527}{1000}.$$

Theorem 22 (Local core). *For every $x \in N$ satisfying (10), either $\text{OPT}_{42}(x) \leq \frac{527}{1000}$, or the unnormalized counts satisfy the four aggregate core inequalities*

$$(13) \quad \frac{36}{25}q \leq r_1 + c_1 \leq \frac{37}{25}q, \quad r_0 + c_0 \leq \frac{9}{100}q, \quad d_1 + a_1 \leq \frac{9}{200}q, \quad (q - d_0) + (q - a_0) \leq \frac{1}{25}q.$$

Both theorems are established by exact branch certificates contained in the ancillary archive (Appendix A, A.1).

Remark 23 (Verification). Two programs sharing no code check both trees, using only exact rational arithmetic and trusting no stored number that they can recompute. The primary checker rebuilds the moment system, every McCormick row, every dual, every split and every empty leaf. The second verifier imports neither the first nor the certificate builders: it derives the 42 rows itself by counting incidence fibres on tori of orders 8, 12, 16, 20, keeping a row only when all label tuples are attained with equal multiplicity (which is the factorization statement) and finding by itself that $\{D, A\}$ is the unique pair that fails to factorize per block, so that the two aggregate rows are derived rather than assumed; it then re-validates the rows as exact identities on random colourings, re-derives the McCormick rows, recomputes every leaf bound from its own partial Lagrangian, and re-proves every emptiness claim by its own interval reasoning. The two programs agree on every count in Table 2, and the second rejects eight deliberately altered copies of the first certificate. File names, commands and test parameters are in Appendix A, A.1.

Proof of Theorems 21 and 22. The exact checks in Remark 23 establish that every internal node makes the declared exhaustive split and that every leaf has the stated type. By

Lemma 18, the stored multipliers of each McCormick-dual leaf certify its bound u . In the first tree, the only exceptional leaves lie in N ; all other leaves are empty or have relaxation value at most $527/1000$. In the second, the only exceptional leaves satisfy (13); all other leaves are empty or have relaxation value at most $527/1000$. Applying Lemma 19 to the two root boxes proves the respective claims. $\square$

Their statistics are:

	Theorem 21	Theorem 22
root box	$[0, 1]^8$	N
nodes	6,929	4,423
split nodes	3,464	2,211
McCormick-dual leaves	3,368	1,639
plaid / core leaves	4	563
exact empty leaves	93	10
maximum depth	22	29
largest reconstructed leaf bound	$\frac{527}{1000} - 1.92 \cdot 10^{-6}$	$\frac{2329117}{4419584} = \frac{527}{1000} - 8.53 \cdot 10^{-7}$

TABLE 2. The two branch certificates. The 93 empty leaves of the first tree split as 1 **rsort**, 9 **csort**, 19 **rcsum**, 21 **dasum**, 43 **budget**; the 10 of the second are all **rcsum**. The four plaid leaves of the first tree all carry the template $(0, s, 0, s, 1, 0, 1, 0)$, i.e. the neighbourhood N ; the 563 core leaves of the second all lie wholly inside (13). The last row shows that both certificates are tight to about $2 \cdot 10^{-6}$.

4.3. The escape comparison.

Lemma 24. $\kappa - \frac{527}{1000} > \frac{1}{125}$ and $\gamma < \frac{1}{2}$. Consequently, for every $q \geq 64$,

$$(14) \quad H(2q) - \frac{527}{1000}q^2 > \frac{q^2}{125} - \frac{q}{2} - \frac{1}{4} > 0.$$

Proof. The inequality $\kappa - \frac{527}{1000} > \frac{1}{125}$ reads $4 - 2\sqrt{3} > \frac{535}{1000} = \frac{107}{200}$, i.e. $693 > 400\sqrt{3}$, i.e. $480249 > 480000$. The inequality $\gamma < \frac{1}{2}$ reads $\frac{1}{\sqrt{3}} > \frac{1}{2}$, i.e. $2 > \sqrt{3}$. The first inequality of (14) is then Lemma 13 combined with these two comparisons. For the second, note that $q \geq 64$ gives

$$\frac{q^2}{125} - \frac{q}{2} - \frac{1}{4} \geq q \left(\frac{64}{125} - \frac{1}{2} \right) - \frac{1}{4} = \frac{3q}{250} - \frac{1}{4} \geq \frac{192}{250} - \frac{1}{4} > 0. \quad \square$$

Combining (9) with Theorems 21, 22 and Lemma 24:

Corollary 25. Let $q \geq 64$ and let a configuration in the canonical chamber have $\min(B, W) \geq H(2q)$. Then its normalized profile lies in N and its unnormalized counts satisfy the aggregate core inequalities (13).

5. THE LOCAL FINISH

Throughout this section $q \geq 130$, the configuration lies in the canonical chamber, its normalized profile lies in the neighbourhood N of (12), and the aggregate core (13) holds. We rename the eight counts to the local coordinates

(15) $u = r_0, \quad R = r_1, \quad v = c_0, \quad C = c_1, \quad e = d_1, \quad g = q - d_0, \quad f = a_1, \quad h = q - a_0,$
so u, v, e, f, g, h are small “defects” and R, C are close to sq . All eight are integers in $[0, q]$. Put

$$(16) \quad \begin{aligned} \sigma &= R + C, & m &= u + v, & \delta &= e + f, & \zeta &= g + h, \\ \alpha &= 2q - \sigma - m, & \beta &= \sigma - q, & Q &= 2ef + 2gh, \\ X &= uv + RC, & Y &= (q - u)(q - C) + (q - R)(q - v). \end{aligned}$$

The core (13) reads

$$(17) \quad \frac{36}{25}q \leq \sigma \leq \frac{37}{25}q, \quad m \leq \frac{9}{100}q, \quad \delta \leq \frac{9}{200}q, \quad \zeta \leq \frac{1}{25}q,$$

and the chamber (10) gives, in these coordinates,

$$(18) \quad u + R \leq v + C, \quad e + h \leq f + g, \quad \sigma + m + \delta - \zeta \leq 2q.$$

Three remarks on notation. First, from here to the end of the section R and C denote the two integer counts of (15); the line sets of Section 2 are not referred to again before Section 7. Likewise the defect sums are written δ, ζ to avoid collision with the antidiagonal counts a_0, a_1 and with the right-hand side $b(x)$ of Section 6. Second, (X, Y) are exactly the two capacities of the H -profile $(r_0, r_1, c_0, c_1) = (u, R, C, v)$ in Definition 1, in which the two column parity counts are *exchanged* relative to the configuration’s own quadruple (u, R, v, C) ; the exchanged quadruple is again admissible in (1), so its capacities are $r_0c_1 + r_1c_0 = uv + RC = X$ and $(q - r_0)(q - c_0) + (q - r_1)(q - c_1) = (q - u)(q - C) + (q - R)(q - v) = Y$, and

$$(19) \quad \min(X, Y) \leq H(2q).$$

Third, α and β are the two “slacks” that the support cuts below will charge against.

5.1. The two support cuts. The proof uses exactly two members of an exactly certified library of 760 support-dual cuts for the moment system (Section 6 and Appendix A); they are entries 609 and 559 of that library, transported to the coordinates (15). A *support dual* is a pair $(\alpha_{\text{cut}}, \lambda)$ of exact rationals such that the corresponding combination of the 42 moment equations dominates $\alpha_{\text{cut}}B + (1 - \alpha_{\text{cut}})W$ columnwise on all 64 atoms; the resulting polynomial inequality then holds for every genuine configuration, with no relaxation and no branching. Cut 609 has $\alpha_{\text{cut}} = 1$ and cut 559 has $\alpha_{\text{cut}} = 0$, so they bound B and W separately:

Lemma 26 (Cuts 609 and 559). *For every configuration,*

$$(20) \quad B \leq F_B := X - \beta\zeta + Q,$$

$$(21) \quad W \leq F_W := Y - \alpha\delta + Q.$$

In particular, writing $T := \min(F_B, F_W)$, we have $\min(B, W) \leq T$.

Both duals are sparse and are printed in Appendix A, A.1, together with the checks of the polynomial identities (20), (21) and of the columnwise support property. Everything below is elementary algebra from (17), (18), (20) and (21).

5.2. Basic estimates and the defect dichotomy.

Lemma 27. *Under (17),*

$$(22) \quad \alpha \geq \frac{43}{100}q, \quad \beta \geq \frac{11}{25}q, \quad \alpha > \delta, \quad \beta > \zeta.$$

Proof. $\alpha = 2q - \sigma - m \geq 2q - \frac{37}{25}q - \frac{9}{100}q = \frac{200-148-9}{100}q = \frac{43}{100}q$, and $\beta = \sigma - q \geq \frac{36}{25}q - q = \frac{11}{25}q$. Finally $\delta \leq \frac{9}{200}q < \frac{43}{100}q \leq \alpha$ and $\zeta \leq \frac{1}{25}q < \frac{11}{25}q \leq \beta$. $\square$

Lemma 28 (Defect dichotomy). *Under (17),*

$$(23) \quad 2Q = 4ef + 4gh \leq \delta^2 + \zeta^2 \leq \alpha\delta + \beta\zeta, \quad \text{hence} \quad Q \leq \max(\alpha\delta, \beta\zeta).$$

Proof. $4ef \leq (e+f)^2 = \delta^2$ and $4gh \leq (g+h)^2 = \zeta^2$. By Lemma 27, $\delta^2 \leq \alpha\delta$ and $\zeta^2 \leq \beta\zeta$. Finally $Q \leq \frac{1}{2}(\alpha\delta + \beta\zeta) \leq \max(\alpha\delta, \beta\zeta)$. $\square$

This dichotomy is the only case distinction the local proof needs. We treat the two orderings of X and Y separately.

5.3. Case $X \leq Y$.

Proposition 29 (First defect case). *If $X \leq Y$ then $T \leq H(2q)$.*

Proof. If $Q \leq \beta\zeta$ then $F_B \leq X$ by (20), so $T \leq X = \min(X, Y) \leq H(2q)$ by (19).

Assume therefore

$$(24) \quad Q > \beta\zeta.$$

By Lemma 28, $Q \leq \alpha\delta$.

Step 1: a bound on F_B . Since $\zeta \leq q/25$,

$$(25) \quad 2gh \leq \frac{\zeta^2}{2} \leq \frac{q\zeta}{50} \leq \beta\zeta,$$

the last step by Lemma 27. Hence

$$(26) \quad F_B = X - \beta\zeta + 2ef + 2gh \leq X + 2ef.$$

Step 2: four switched profiles. Consider the four H -profiles

$$(27) \quad (r_0, r_1, c_0, c_1) = (u+x, R, C, v+y), \quad (x, y) \in \{(e, 2f), (2e, f), (f, 2e), (2f, e)\}.$$

All coordinates lie in $[0, q]$: inside N we have $u, v \leq q/10$ and $e, f \leq q/10$, so the largest modified coordinate is at most $q/10 + 2q/10 = 3q/10$. The black capacity of (27) is

$$(28) \quad (u+x)(v+y) + RC = X + uy + vx + xy \geq X + xy = X + 2ef,$$

since $u, v, x, y \geq 0$ and each of the four choices in (27) has $xy = 2ef$. Combining with (26), each of the four profiles has black capacity at least F_B .

The white capacity of (27) is $(q-u-x)(q-C) + (q-R)(q-v-y) = Y - (x(q-C) + y(q-R))$, so the *white loss* is $x(q-C) + y(q-R)$. Put $L = 2q - \sigma$. Over the four choices in (27) the values of x sum to $3(e+f) = 3\delta$ and so do the values of y ; hence the average white loss is exactly

$$(29) \quad \frac{3\delta}{4}((q-C) + (q-R)) = \frac{3}{4}\delta L.$$

Step 3: the averaged loss is affordable. We claim

$$(30) \quad Q \leq \delta\left(\frac{L}{4} - m\right).$$

First, by (17),

$$(31) \quad \frac{L}{4} - m \geq \frac{1}{4} \left(2 - \frac{37}{25} \right) q - \frac{9}{100} q = \frac{13}{100} q - \frac{9}{100} q = \frac{q}{25}.$$

Second, from (24), Lemma 27 and (25),

$$2ef = Q - 2gh > \beta\zeta - \frac{q\zeta}{50} \geq \frac{11}{25} q\zeta - \frac{q\zeta}{50} = \frac{21}{50} q\zeta.$$

Since $2ef \leq \delta^2/2$, this gives $\zeta < \frac{25\delta^2}{21q}$, and therefore

$$(32) \quad Q \leq \frac{\delta^2 + \zeta^2}{2} \leq \frac{\delta^2}{2} + \frac{q\zeta}{50} < \frac{\delta^2}{2} + \frac{q}{50} \cdot \frac{25\delta^2}{21q} = \frac{11}{21} \delta^2 \leq \frac{11}{21} \cdot \frac{9q}{200} \delta = \frac{33}{1400} \delta q < \frac{1}{25} \delta q,$$

using $\zeta^2/2 \leq q\zeta/50$ (valid as $\zeta \leq q/25$) and $\delta \leq 9q/200$. Comparing (31) and (32) proves (30).

Step 4: conclusion. Since $\alpha = L - m$, inequality (30) is equivalent to

$$(33) \quad \frac{3}{4} \delta L \leq \delta(L - m) - Q = \alpha\delta - Q.$$

By (29) at least one of the four profiles has white loss at most $\frac{3}{4}\delta L$, hence white capacity at least $Y - (\alpha\delta - Q) = F_W$. By (28) and (26) its black capacity is at least F_B . That profile therefore has H -value at least $\min(F_B, F_W) = T$, and so $T \leq H(2q)$. $\square$

5.4. Case $Y \leq X$.

Proposition 30 (Second defect case). *If $Y \leq X$ and $q \geq 130$, then $T \leq H(2q)$.*

Proof. If $Q \leq \alpha\delta$ then $F_W \leq Y$ by (21), so $T \leq Y = \min(X, Y) \leq H(2q)$ by (19).

Assume therefore

$$(34) \quad Q > \alpha\delta.$$

By Lemma 28, $Q \leq \beta\zeta$.

Step 1: the integrality step. We claim $gh > 0$. If $gh = 0$, then (34) forces $\delta > 0$: otherwise Lemma 27 gives $e = f = 0$, and hence $Q = 0$, contradicting $Q > \alpha\delta = 0$. Using $\delta \leq \frac{9}{200}q$ and Lemma 27, we then have

$$Q = 2ef \leq \frac{\delta^2}{2} \leq \frac{9q}{400} \delta < \frac{43}{100} q \delta \leq \alpha\delta,$$

contradicting (34). Since g and h are integer counts, $g \geq 1$ and $h \geq 1$, so

$$(35) \quad \zeta = g + h \geq 2.$$

This is the discrete step in the local defect argument, where the continuous relaxation is insufficient (Remark 31). Integrality is also used in the construction of H and the finite-range proof.

Step 2: averaging the two cuts. By (20) and (21),

$$(36) \quad T \leq \frac{F_B + F_W}{2} = \frac{X + Y}{2} - \frac{\alpha\delta + \beta\zeta - 2Q}{2}.$$

By (23),

$$(37) \quad \alpha\delta + \beta\zeta - 2Q \geq \alpha\delta + \beta\zeta - (\delta^2 + \zeta^2) = \delta(\alpha - \delta) + \zeta(\beta - \zeta) \geq \zeta(\beta - \zeta),$$

since $\delta(\alpha - \delta) \geq 0$ by Lemma 27. On the range $2 \leq \zeta \leq q/25$ the function $\zeta \mapsto \zeta(\beta - \zeta)$ is increasing, because its derivative $\beta - 2\zeta \geq \frac{11}{25}q - \frac{2}{25}q = \frac{9}{25}q > 0$. Hence by (35),

$$(38) \quad \frac{\alpha\delta + \beta\zeta - 2Q}{2} \geq \frac{2(\beta - 2)}{2} = \beta - 2.$$

Step 3: the balanced-plaid baseline. We claim

$$(39) \quad \frac{X + Y}{2} \leq \kappa q^2.$$

Normalize (u, R, C, v) by q and put, as in Lemma 12,

$$\rho = \frac{u + R}{q}, \quad \chi = \frac{v + C}{q}, \quad \xi = 1 - \rho, \quad \eta = 1 - \chi, \quad z = \left(\frac{u}{q} - \frac{R}{q}\right)\left(\frac{C}{q} - \frac{v}{q}\right).$$

The neighbourhood N gives $\rho \leq \frac{1}{10} + \frac{7}{8} < 1$ and $\chi < 1$, so $\xi, \eta > 0$. By (3),

$$\frac{X - Y}{q^2} = \rho + \chi - 2 - z, \quad |z| \leq \rho\chi, \quad \frac{X + Y}{q^2} = 1 + \xi\eta.$$

The assumption $X \geq Y$ forces $-\rho\chi \leq z \leq \rho + \chi - 2$, whence $2 - \rho - \chi \leq \rho\chi$, i.e. $2\xi + 2\eta - \xi\eta \leq 1$. With $w = \sqrt{\xi\eta}$, AM–GM gives $4w - w^2 \leq 1$, so $w \leq 2 - \sqrt{3}$ and

$$\frac{X + Y}{2q^2} = \frac{1 + w^2}{2} \leq \frac{1 + (2 - \sqrt{3})^2}{2} = 4 - 2\sqrt{3} = \kappa,$$

which is (39).

Step 4: conclusion. Combining (36), (38), (39) and $\beta \geq \frac{11}{25}q$,

$$(40) \quad T \leq \kappa q^2 - \frac{11}{25}q + 2.$$

By Lemma 13, $H(2q) \geq \kappa q^2 - \gamma q - \frac{1}{4}$, so (40) gives $T \leq H(2q)$ as soon as

$$(41) \quad \left(\frac{11}{25} - \gamma\right)q \geq \frac{9}{4}.$$

Now $\frac{11}{25} - \gamma = \frac{1}{\sqrt{3}} - \frac{14}{25}$, so (41) at $q = 130$ reads $\frac{130}{\sqrt{3}} - \frac{14 \cdot 130}{25} \geq \frac{9}{4}$, i.e. $\frac{130}{\sqrt{3}} \geq \frac{1501}{20}$, whose square is the integer comparison

$$6,760,000 \geq 6,759,003.$$

This holds, and the left-hand side of (41) is increasing in q (the coefficient $\frac{1}{\sqrt{3}} - \frac{14}{25} = 0.01735\dots$ is positive), so (41) holds for every $q \geq 130$. $\square$

Remark 31. The margin in (41) is $0.01735\dots$ per unit of q against a fixed deficit of $9/4$, which is why the threshold sits at $q = 130$ and not lower. The particular continuous relaxation used before the integrality step is not sharp at $n = 18$ and $n = 20$: it gives only $t \leq 43$ and $t \leq 53$, against the exact values 42 and 51. The integrality step (35) — and, in the finite range, the integer cut envelope of Section 6 — closes the remaining one- and two-unit gaps.

5.5. Proof of the upper bound for $q \geq 130$.

Theorem 32 (Large even orders). $t(2q) \leq H(2q)$ for every $q \geq 130$.

Proof. Suppose $t(2q) \geq H(2q) + 1$. By Lemma 6 there is a configuration with $\min(B, W) \geq H(2q) + 1$, and by Lemma 17 we may assume it lies in the canonical chamber. Since $q \geq 130 \geq 64$, Corollary 25 places its normalized profile in N and its counts in the aggregate core (13). Lemma 26 gives $\min(B, W) \leq T$, and Propositions 29 and 30 — which between them cover both orderings of X and Y — give $T \leq H(2q)$. Hence $\min(B, W) \leq H(2q)$, a contradiction. $\square$

Explicitly, a canonical profile falls into exactly one of three branches: outside N , where Theorem 21 and Lemma 24 give $\min(B, W) < H(2q)$; inside N but outside the aggregate core, where Theorem 22 and Lemma 24 give the same; and inside the core, where Section 5 gives $\min(B, W) \leq H(2q)$.

6. THE FINITE RANGE $q \leq 129$

6.1. The certified cut library. The support cuts used in Section 5 belong to a library of 760. A cut is a pair (θ, λ) with $\theta \in \mathbb{Q} \cap [0, 1]$ and $\lambda \in \mathbb{Q}^{42}$ such that, on each of the 64 atom columns,

$$(42) \quad \theta \mathbf{1}[\varepsilon = (1, 1, 1, 1)] + (1 - \theta) \mathbf{1}[\varepsilon = (0, 0, 0, 0)] \leq (\lambda^\top E)_{i,j,\varepsilon}$$

for every block (i, j) and atom $\varepsilon \in \{0, 1\}^4$, where E is the 42×64 atom-variable coefficient matrix of the moment equations. Its columns are indexed by (i, j, ε) .

Lemma 33 (Support-cut inference). *If (42) holds, then every genuine configuration satisfies*

$$\min(B, W) \leq \theta B + (1 - \theta)W \leq q^2 \lambda^\top b(x),$$

where $b(x)$ is the moment right-hand side at its outer profile.

Proof. The first inequality holds because the middle expression is a convex combination of B and W . Multiply the columnwise inequalities (42) by the nonnegative atom counts and sum. The moment equations replace Ep by $b(x)$. $\square$

The library is stored in two files: 76 cuts as polynomials in the outer counts and 684 as rational duals (θ, λ) . An exact standard-library verifier reconstructs the 42 moment rows, checks the 64 column inequalities and the polynomial identity for every stored dual, and recovers a rational witness λ for each of the 76 polynomial cuts by exact simplex; it verifies all 760 cuts (Appendix A, A.1).

6.2. The cut envelope at a fixed order. For fixed q , evaluate each cut P at integer outer counts. Let s be the least common multiple of 2 and of all coefficient denominators of the library; in the released library $s = 12$. The engine stores the integer coefficients of sP , scales constants by q^2 and linear terms by q , and performs every comparison as the exact integer test $sP(x) \leq sH(2q) + s/2$, so the threshold $H(2q) + \frac{1}{2}$ is represented without rounding. The minimum of the cuts is the *cut envelope*. The engine explores boxes $[l, u] \subseteq \{0, \dots, q\}^8$ starting from the full cube, prunes and tightens by the four chamber and the budget inequalities, and discharges a box when one cut is certified to stay at most $H(2q) + \frac{1}{2}$ on it, by a term-wise interval bound or by an exact corner maximum; one-point boxes are discharged by direct evaluation. The following lemma records why each of these operations is sound.

Lemma 34 (Box-discharge soundness). *Throughout the search $0 \leq l \leq u \leq q$ coordinatewise, and:*

- (i) *Propagation. Each of the five canonical inequalities has coefficients in $\{0, \pm 1\}$. A box is pruned only when the minimum of a row over the whole box, evaluated coordinatewise at endpoints, exceeds its right-hand side; and a bound is tightened only to a value implied by the row for every point of the box satisfying it. Hence no integer point satisfying the inequalities is ever discarded. Since the root box is the full cube, every canonical profile is covered; points violating the chamber are covered as well, which is harmless.*

- (ii) *Multi-affinity.* Every cut polynomial is multi-affine: its quadratic monomials use only products of two distinct coordinates (the 22 admissible pairs), never a square, so it is affine in each coordinate separately. The engine rejects any other input shape at parse time.
- (iii) *Corner maximum.* A multi-affine function attains its maximum over a real box at a vertex. The corner test enumerates all vertices in the coordinates that occur in the cut and are undetermined in the box; if none exceeds the threshold, the cut is at most the threshold on the whole real box, in particular at every integer point.
- (iv) *Interval bound.* Since $0 \leq l \leq x \leq u$, each bilinear term $c_{ij}x_ix_j$ is at most $c_{ij}u_iu_j$ for $c_{ij} > 0$ and at most $c_{ij}l_il_j$ for $c_{ij} < 0$, and each linear term is bounded at its matching endpoint; the term-wise sum is therefore a valid upper bound for the cut on the box.
- (v) *Splitting.* A split at $c = \lfloor (l_i + u_i)/2 \rfloor$ into $x_i \leq c$ and $x_i \geq c + 1$ partitions the integer points of the parent box.
- (vi) *No overflow.* All engine arithmetic is signed 64-bit. Let S be the maximum over the library of the sum of the absolute values of all scaled coefficients of one cut; the driver recomputes S before every run and aborts unless $Sq^2 < 2^{63}$. In the released library $S = 360$, and every intermediate quantity of the engine is bounded in absolute value by $2Sq^2$, which even at the engine's input cap $q \leq 1000$ is $7.2 \cdot 10^8 \ll 2^{63}$; every comparison is therefore exact.

Proof. (i) The minimum of a row over the box is attained coordinatewise at endpoints because the row is linear; if it exceeds the right-hand side, no point of the box satisfies the row. Otherwise, for a coordinate with coefficient $+1$, every satisfying point obeys $x_i \leq \text{rhs} - (\text{minimum of the remaining terms over the box})$, which is an integer since all coefficients are ± 1 ; tightening u_i to it (and l_i symmetrically for coefficient -1) keeps every satisfying point. (ii) is structural in the stored cut files and is reverified by the audits. (iii) Fixing all but one coordinate, a multi-affine function is affine, hence maximized at an endpoint; induct on the dimension. (iv) All coordinates are nonnegative because l only ever increases from 0, so the displayed per-term extrema are exact. (v) is immediate. (vi) For $x \in [0, q]^8$ every term of sP is bounded by its absolute coefficient times q^2 , so partial sums are bounded by Sq^2 ; the corner test's incremental updates combine at most two such sums. The stated values of s and S are recomputed by the release audit. $\square$

Lemma 35 (Integer-box inference). *Suppose such a search ends with no survivor and every nonempty terminal box has a cut bounded above by $H(2q) + \frac{1}{2}$ throughout the box. Then $t(2q) \leq H(2q)$.*

Proof. Splits partition the integer points of their parent boxes by Lemma 34(v). Pruning removes only boxes containing no canonical integer profile by Lemma 34(i). Induction on the tree therefore places every canonical integer profile in a discharged terminal box. At this profile the discharging cut is at most $H(2q) + \frac{1}{2}$, by Lemma 34(iii), (iv), or direct evaluation for a one-point box. Lemma 33 then bounds $\min(B, W)$ by the same quantity. Since $\min(B, W)$ and $H(2q)$ are integers, we obtain

$$\min(B, W) \leq H(2q). \quad \square$$

Proposition 36. *For*

$$q \in \{3, 5\} \cup \{10, 11, \dots, 129\},$$

the cut envelope is at most $H(2q) + \frac{1}{2}$ throughout the canonical integer domain. Consequently $t(2q) = H(2q)$ for these orders.

Proof. We ran the C++17 exact box engine of the ancillary archive on all 122 listed orders. Every search exhausted its stack with no survivor, visiting 13,379,092 nodes in total; every discharge used one of the 76 polynomial cuts, and the fallback sweep of the full library at one-point boxes was never invoked. The release verifier recompiles that engine, repeats every search, recomputes $H(2q)$ and compares every proof-relevant output field; thus it re-establishes the domain coverage required by Lemma 35.

The archived per-order JSON files bind the cut inputs and record witnesses, status fields and aggregate statistics, but are compact run summaries, not node-by-node tree certificates. Their checker audits those data and the full-tree accounting without claiming geometric coverage. As a separate implementation check, a Python engine was also run on every order: it produced the archived summaries for $q \leq 60$, which the C++ engine reproduces, and for $61 \leq q \leq 129$ its non-runtime envelope outputs agree exactly with the C++ outputs. The two engines implement the same box algorithm, so this agreement is corroborative rather than a conceptually independent proof method. Lemma 35 gives the upper bound, and Proposition 10 gives equality. $\square$

The orders $q = 1, 2$ are elementary. The four remaining envelope exceptions $q = 4, 6, 7, 9$ are direct finite searches, while $q = 8$ is covered by the union-domain verification of $n = 16$. They are proved together with the odd searches in Section 7. Independently of both engines, the Lean development described in Appendix A, A.0, checks complete action trees for every $3 \leq q \leq 129$, including these five orders, inside a proof assistant.

7. THE EXHAUSTIVE FINITE SEARCHES

On an odd torus the map φ of Lemma 7 is bijective, so label parity carries no geometric information. We now give the finite arguments for $t(15)$ and $t(17)$ and for the five even orders not covered by Proposition 36.

7.1. The common search reduction. For odd n , write

$$(s_0, s_1, s_2, s_3) = (|R|, |C|, |D|, |A|), \quad S = \sum_i s_i.$$

If a target M is feasible, Lemmas 8, 7 and 9 imply

$$(43) \quad S \leq 2n,$$

$$(44) \quad s_i s_j \geq M, \quad (n - s_i)(n - s_j) \geq M \quad (i < j),$$

$$(45) \quad n^2 - n(x + y + z) + xy + xz + yz \geq 2M \quad \text{for each triple } x, y, z.$$

These necessary conditions give a finite profile list. We quotient only by explicit affine symmetries: transpose, $(r, c) \mapsto (r, -c)$, and, for odd n , $(r, c) \mapsto (r + c, r - c)$. No classification of the full affine symmetry group is used.

For each surviving profile, fix two line families, say R, C , up to a subgroup of their affine stabilizer. For an enumerated set A , define

$$\begin{aligned} p_d &= \#\{(r, c) : r \in R, c \in C, r - c = d, r + c \in A\}, \\ q_d &= \#\{(r, c) : r \notin R, c \notin C, r - c = d, r + c \notin A\}, \quad Q_0 = \sum_d q_d. \end{aligned}$$

For any D of its prescribed size,

$$(46) \quad |B| = \sum_{d \in D} p_d, \quad |W| = Q_0 - \sum_{d \in D} q_d.$$

Thus the last family is an exact fixed-cardinality, two-constraint subset problem.

Lemma 37 (Exhaustive-search inference). *Assume that the profile filter uses only necessary conditions, the fixed-pair list meets every orbit of a subgroup preserving the searched profile domain, every admissible third-family set is enumerated, and (46) is decided exactly for the last family. If every listed profile is infeasible, then the target M is infeasible.*

Proof. Any feasible line colouring has a profile in the filtered list. A permitted affine symmetry sends its fixed pair to an enumerated representative and transports the remaining line sets within the searched domain. Its third set occurs in the enumeration, and its last set satisfies (46); an exact decision procedure must therefore find it. $\square$

7.2. The value $t(15)$. The line sets

$$\begin{aligned} R &= \{5, 7, 8, 11, 12, 14\}, & C &= \{0, 3, 4, 7, 9, 10, 12, 13\}, \\ D &= \{0, 1, 4, 8, 12, 13, 14\}, & A &= \{1, 2, 3, 9, 10, 12, 14\} \end{aligned}$$

have $|B| = 20$ and $|W| = 22$, proving the lower bound.

At target $M = 21$, conditions (43)–(45) leave 247 canonical profiles. Their counts in the strata $S = 20, \dots, 30$ are

$$1, 1, 4, 6, 13, 18, 29, 37, 47, 52, 39.$$

For each profile, the fixed-pair subgroup is generated by independent translations, common multiplication by a unit, transpose when the fixed sizes agree, and $(R, C) \mapsto (R, -C)$ only when the two completion sizes agree. The last restriction ensures that the reflection does not exchange two distinct ordered profiles. A separate audit regenerates a canonical identifier for every orbit and checks all 43 fixed-pair orbit types by Burnside’s lemma.

The first enumerator decides (46) by depth-first search, using necessary cardinality bounds on each suffix. An independent enumerator chooses a different fixed pair on 19 profiles and uses an exact 7 + 8 meet-in-the-middle Pareto table. Their respective case totals are

$$6,074,753,568 \quad \text{and} \quad 6,241,793,402.$$

Both return UNSAT on all 247 profiles; the complete second result file is checked for exact profile coverage and unique UNSAT records.

Theorem 38. $t(15) = 20$.

Proof. The displayed line sets give $t(15) \geq 20$. The profile generation, orbit audit, complete third-family enumeration and the two exact last-family decisions satisfy Lemma 37 at target 21. Hence $t(15) \leq 20$. $\square$

7.3. A union-domain verification at $n = 16$. Although Theorem 2 already gives $t(16) = 32$, a direct search provides an independent small-order proof. Let $E, O \subset \mathbb{Z}/16\mathbb{Z}$ be the even and odd residue classes and use profiles

$$(r, c, d_0, d_1, a_0, a_1) = (|R|, |C|, |D \cap E|, |D \cap O|, |A \cap E|, |A \cap O|).$$

The budget, pair bounds and triple identity leave exactly 1,898 ordered profiles at target 33.

The proof search groups these profiles only by (r, c) , after transposing so that $r \leq c$, and searches the *union* of all associated completion profiles. This union is invariant under simultaneous parity exchange and under exchanging D with A . Hence independent translations, common multiplication by an odd unit, $(R, C) \mapsto (R, -C)$, and transpose when $r = c$ act within the searched domain; no parity-orientation transfer is needed. Fixed-pair representatives are selected by a lexicographic canonical-form test on translation necklaces.

The enumerator scans every admissible A -mask. Only the two necessary row–column–antidiagonal triple inequalities are used as a scalar prefilter; for every survivor it scans every admissible D -mask literally. The exact totals are

fixed-pair representatives	159,551
A -masks tested	3,226,530,570
A -masks surviving the scalar filter	2,951
diagonal-cardinality cases	29,768
D -masks tested	38,830,322.

None is feasible. A separately written Python audit regenerates the profile domain, proves its closure under the three domain symmetries, computes the seven pair-orbit counts by Burnside’s lemma, regenerates every representative list and hash, and checks the certificate totals. It prints UNION_AUDIT_OK. Two further programs (677 jobs), which share the search chassis but use different completion kernels, reproduce the same upper bound.

Proposition 39. *The union-domain search certifies that no $33 + 33$ placement exists on the 16×16 torus.*

Proof. The 1,898 profiles contain every feasible colouring by the necessary conditions. The union domain makes every quotient symmetry profile preserving, and the audited representative list meets every fixed-pair orbit. All third-family masks and then all admissible last-family masks are tested, so Lemma 37 applies. $\square$

7.4. The remaining orders.

Lemma 40 (Parity-refined even sweeps). *Let $n = 2h$, and suppose a line colouring has target M . With $P_\varepsilon = \{x \in \mathbb{Z}/n\mathbb{Z} : x \equiv \varepsilon \pmod{2}\}$, give it the profile*

$$(r, c, d_0, d_1, a_0, a_1) = (|R|, |C|, |D \cap P_0|, |D \cap P_1|, |A \cap P_0|, |A \cap P_1|).$$

Put $s_R = r$, $s_C = c$, $s_D = d_0 + d_1$, $s_A = a_0 + a_1$. For distinct families X, Y , define

$$J_{DA} = 2(d_0 a_0 + d_1 a_1),$$

$$\bar{J}_{DA} = 2((h - d_0)(h - a_0) + (h - d_1)(h - a_1)),$$

$$J_{XY} = s_X s_Y, \quad \bar{J}_{XY} = (n - s_X)(n - s_Y) \quad (\{X, Y\} \neq \{D, A\}).$$

After possibly complementing all four line sets, every feasible profile satisfies

$$(47) \quad \sum_X s_X \leq 2n, \quad J_{XY}, \bar{J}_{XY} \geq M, \quad n^2 - n(s_X + s_Y + s_Z) + J_{XY} + J_{XZ} + J_{YZ} \geq 2M$$

for every pair and every triple of distinct families.

It is enough to retain one profile from each orbit generated by exchanging R, C , exchanging D, A , simultaneously swapping d_0, d_1 and a_0, a_1 , and, on the boundary $\sum_X s_X = 2n$, complementation. For each retained profile, choose an R, C and D, A orientation and search both simultaneous parity orientations unless they coincide. Fixing R, C , one may quotient by independent translations and common multiplication by a unit, as well as transpose when

$r = c$ and $(R, C) \mapsto (R, -C)$ when $(d_0, d_1) = (a_0, a_1)$. Exhausting the resulting pair representatives, all A with the prescribed parity sizes, and the exact parity-constrained D -completion problem exhausts the target M .

Proof. The first inequality in (47) is Lemma 8. Lemma 7 gives J_{XY} and $\bar{J}_{XY}$ as the black and white pair-intersection counts; the exceptional formulas for D, A use its two-or-zero fibres. Since B and W lie in the corresponding intersections, the pair inequalities follow. Applying Lemma 9 to each triple gives the last inequality.

Transpose, $(r, c) \mapsto (r, -c)$, and translation by one row induce the three profile exchanges above; complementation swaps the armies. An independent translation of R, C shifts the D, A labels by amounts of the same parity, so it either preserves both parity orientations or exchanges both. Every unit modulo $2h$ is odd and preserves parity. Transpose and the relative-sign map preserve a job under precisely the stated size conditions. Thus the union of the paired orientations is stable under the fixed-pair quotient and meets every feasible pair orbit. For each representative, the third-family enumeration uses only necessary suffix bounds. Every surviving A is passed to the exact subset problem (46), now with the two constraints $|D \cap P_\varepsilon| = d_\varepsilon$. Lemma 37 then gives the asserted coverage. $\square$

The same profile-orbit-completion programme was implemented for arbitrary $n \leq 18$, using Lemma 40 at even orders. Its independent audit regenerates every worklist, the Burnside orbit counts and the complete candidate-space arithmetic. The upper campaigns needed here are:

n	target	jobs	third-family cases
8	9	6	720
12	19	100	5,111,848
14	26	76	70,259,455
17	29	316	157,158,165,660
18	43	259	228,644,005,008

Every job is UNSAT. For $n = 17$, a second implementation with a different quotient and enumeration order also returns UNSAT on all 316 jobs, after 158,802,287,408 logical third-family cases.

At $n = 17$ the line sets

$$\begin{aligned} R &= \{0, 1, 2, 4, 5, 7, 8, 9\}, & C &= \{0, 1, 2, 4, 5, 7, 8, 9, 13\}, \\ D &= \{2, 3, 4, 7, 10, 13, 14, 15\}, & A &= \{2, 5, 6, 7, 9, 11, 12, 13, 16\} \end{aligned}$$

have $|B| = |W| = 28$.

Theorem 41. $t(17) = 28$.

Proof. The displayed sets give the lower bound. The audited 316-job campaign meets the hypotheses of Lemma 37 at target 29, giving the upper bound. $\square$

Proposition 42. For $q \in \{1, 2, 4, 6, 7, 8, 9\}$, one has $t(2q) = H(2q)$.

Proof. For $q = 1$, any two distinct cells of the 2×2 torus share a row, a column or a diagonal, so no black-white pair coexists and $t(2) = 0 = H(2)$. For $q = 2$, suppose $\min(B, W) \geq 3$ for some line colouring of the 4×4 torus. For each of the pairs (R, C) , (R, D) , (C, D) and (R, A) — none of which is the pair $\{D, A\}$ — Lemma 7 matches the label pairs bijectively with cells, so with x, y the two black sizes, $xy \geq 3$ and $(4-x)(4-y) \geq 3$; the only integer solutions have $x + y = 4$. The first three pairs force $|R| = |C| = |D| = 2$, and the fourth then gives $|A| = 2$.

The triple identity (2) for R, C, D now reads $U + U' = 16 - 4 \cdot 6 + 3 \cdot 4 = 4 < 6 \leq |B| + |W|$, a contradiction. Hence $t(4) \leq 2 = H(4)$. For $q = 4, 6, 7, 9$, the first, second, third and fifth rows of the campaign table exclude $H(2q) + 1$ queens of each colour. Proposition 39 does the same for $q = 8$. Proposition 10 supplies every lower bound. $\square$

Proof of Theorems 2 and 5. Proposition 10 gives $t(2q) \geq H(2q)$. The reverse inequality is Theorem 32 for $q \geq 130$, Proposition 36 for $q \in \{3, 5\} \cup \{10, \dots, 129\}$, and Proposition 42 for the remaining q . Theorems 38 and 41 give Theorem 5. $\square$

Proof of the corollaries. Corollary 3 follows from Theorem 2 and Lemmas 12 and 13. The displayed values follow by evaluating the finite maximum (1). $\square$

The odd construction below explains some of the sensitivity to divisors, but does not suggest an odd analogue of Theorem 2.

Proposition 43 (Divisor lifting). *If $d \mid n$ then $t(n) \geq (n/d)^2 t(d)$.*

Proof. Let (R, C, D, A) be an optimal configuration on the $d \times d$ torus and let $\pi : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/d\mathbb{Z}$ be reduction. Take $(\pi^{-1}R, \pi^{-1}C, \pi^{-1}D, \pi^{-1}A)$. A cell (r, c) of the n -torus is black-homogeneous iff $(\pi r, \pi c)$ is, since $\pi(r - c) = \pi r - \pi c$ and $\pi(r + c) = \pi r + \pi c$; and each cell of the d -torus has exactly $(n/d)^2$ preimages. So both armies scale by $(n/d)^2$, and Lemma 6 applies. $\square$

For example, $t(39) \geq 144$, $t(45) \geq 180$ and $t(51) \geq 252$. The lift from proper divisors is not generally sharp:

$$\max_{\substack{d \mid 9 \\ d < 9}} (9/d)^2 t(d) = 0 < 7 = t(9), \quad \max_{\substack{d \mid 15 \\ d < 15}} (15/d)^2 t(d) = 18 < 20 = t(15).$$

The odd asymptotics remain open; the upper bound $t(n) \leq n^2/8$ is [2, Thm. 1.4].

8. USE OF AI TOOLS

Large-language-model tools assisted with conjecture generation, drafting, code generation and source checking. The author checked the final mathematical arguments and proof-critical outputs and takes responsibility for the paper. Agreement among AI-assisted implementations is treated as an error-detection measure, not as external scholarly reproduction.

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APPENDIX A. VERIFICATION AND REPRODUCTION

The complete ancillary archive is bound by SHA256SUMS. The archive is packaged without platform metadata files, and the manifest check of the release verifier is two-sided: it fails on any missing, altered or unlisted file, ignoring only byte-code caches. An immutable public snapshot is available at

<https://github.com/jphme/math-problems/tree/e3f823ca5f382687eb52486fdf048476cefcb85b/peaceable-queens-even-torus>.

The following command, run from `anc/`, replays the exact proof objects, reruns the complete finite cut envelope and audits the frozen outputs of the larger searches:

```
uv run --isolated --python 3.14.6 \
  --with sympy==1.14.0 --with mpmath==1.3.0 \
  python verify_bundle.py
```

It ends with

PEACEABLE_QUEENS_RELEASE_BUNDLE_OK.

The recorded environment is CPython 3.14.6 and SymPy 1.14.0; all other Python checks use only the standard library. The sources use C++17 or C++20 as specified by their build commands and were rebuilt with Apple Clang 21.0.0; the union-domain enumerator was also reproduced with GCC 14 and Clang 17. No floating-point comparison affects a proof verdict.

A.0 Lean formalization of the even theorem. A Lean 4 development formalizes Theorem 2: the line-colouring reduction, the parity construction, the 42 moment equations, both branch trees and the integer local finish. Its finite range is proved by a route different from Section 6: one complete action tree per order, checked node by node inside Lean, covers every $3 \leq q \leq 129$, including the five sweep exceptions of Proposition 42. These trees use only the 76 polynomial cuts of the library, each with a kernel-checked support witness, and they are not a replay of the C++ engine. The two base boards use translations and small kernel decisions.

The sources, documentation and reproduction instructions are published next to this paper, in the `lean/` directory of

<https://github.com/jphme/math-problems/tree/main/peaceable-queens-even-torus>.

The `README.md` of that directory records the revision of the development that accompanies each version of this paper. Lean’s `H q` denotes this paper’s $H(2q)$. `PeaceableQueens/Main.lean` exports the unconditional upper bound and equality; `PeaceableQueens/Corollaries.lean` states Corollaries 3 and 4 explicitly. The separate black and white inequalities in Lemma 26 and the arbitrary-set counting in Proposition 10 are also exported.

`lean/FORMAL_PROOF.md` gives the proof as obligations S1–S66; `lean/PAPER_CORRESPONDENCE.md` maps every numbered result of this paper to its Lean statement and states the coverage limits; the concise guide is `lean/FORMAL_PROOF_GUIDE.pdf`. `lean/VERIFICATION_FILES.txt` lists the exact main-theorem source closure and `lean/COROLLARIES_FILES.txt` adds the value table and analytic corollaries; both include the pinned project files. `lean/VERIFICATION.md` records the complete build and the endpoint axiom audit. The repository’s continuous integration rebuilds the hand-written soundness modules and a sample of certificate modules on every change; the complete closure is too large for hosted runners and is verified offline as recorded there. The companion `lean/scripts/check_paper_values.cpp` reproduces Table 1 and the numerical comparisons in Remark 11, including the 200-order histogram, using exact integers and a small direct-enumeration cross-check.

The proof uses Lean 4.32.2 and Mathlib 4.32.2. Concrete branch batches and finite-tree acceptance use `native_decide`, which additionally trusts Lean’s compiler, runtime and evaluated library implementations. The two main endpoints have three standard axioms and 483 native computation axioms: 356 branch batches and 127 finite acceptances. Checker soundness, support inequalities, small boards and analytic estimates use only the standard axioms.

The formalization bounds rational feasible points and integer boxes. The real optimum in Definition 16 and its attainment, odd-order theorems, divisor lifting, plaid calculations and

the open plaid-plus-two conjecture are outside its scope. The ancillary verification below is a separate route through the paper’s own proof objects and computations.

A.1 Exact proof objects. The files `even_c527_sym_mc.json.gz` and `even_local_core_mc.json.gz` contain the two branch trees. The primary checker `verify_even_branch_certificate.py` reconstructs every split, moment row, McCormick inequality and rational leaf dual and ends in `ALL_EVEN_BRANCH_CERTIFICATES_OK`. The second verifier `verify_even_finite_independent.py` imports neither the checker nor the certificate builders; it derives the 42 moment rows from torus incidence fibres at $n = 8, 12, 16, 20$ and re-validates them on 200 random colourings at each of $n = 12, 16, 20, 28$ (8,400 equation checks) before rechecking both trees (Remark 23).

The files `new_dual_cuts.json` and `benders_cuts.json` together contain the 760 support cuts. `verify_support_duals.py` verifies all of them exactly and recovers rational duals for the 76 polynomial cuts of `benders_cuts.json` independently. The two duals used in Section 5 are stored separately in `even_finite_support_cuts.json`, byte-identical to entries 609 and 559 of `new_dual_cuts.json`; `verify_even_finite_algebra.py` recomputes both polynomial identities and the columnwise support property, and `verify_even_finite_prose.py` checks the displayed algebra and exact constants of that section.

For hand verification, both duals are sparse. Index the 42 rows as in Lemma 14: for each block (i, j) the normalization row, the four marginal rows $E_{ij}[\varepsilon_R]$, $E_{ij}[\varepsilon_C]$, $E_{ij}[\varepsilon_D]$, $E_{ij}[\varepsilon_A]$ and the five pair rows in the order of (6), followed by the two aggregate rows (7). Cut 609 ($\theta = 1$) has multiplier 1 on the normalization row, -1 on the ε_D and ε_A rows and 1 on the $\varepsilon_{R \in C}$ row of block $(0, 0)$; 3 on the normalization row, -2 on each of the four marginal rows and 1 on each of the five pair rows of block $(1, 1)$; 1 on both aggregate rows; and 0 on the remaining 26 rows. Cut 559 ($\theta = 0$) has 1 on the normalization row and -1 on the ε_D and ε_A rows of each of the blocks $(0, 0)$ and $(1, 1)$; 1 on the normalization row, -1 on each marginal row and 1 on each pair row of each of the blocks $(0, 1)$ and $(1, 0)$; 1 on both aggregate rows; and 0 on the remaining 14 rows. Substituting the right-hand sides of Lemma 14 and multiplying by q^2 gives $q^2 \lambda^T b(x) = F_B$ for cut 609 and F_W for cut 559 in the coordinates (15), a short expansion; the 64 columnwise inequalities (42) are a finite check, carried out in the Lean kernel in `lean/PeaceableQueens/UpperBound/SupportCuts.lean`.

A.2 Finite computations. The directories `records_q003_q020/` and `records_q021_q129/` contain compact summaries of the 122 ladder runs. They do not contain the split nodes or terminal boxes. `check_envelope_records.py` recomputes H and checks the order set, schemas, source and input hashes, thresholds, witnesses and aggregate node/leaf accounting. Full coverage is re-established by

```
python replay_finite_envelope.py
```

This compiles `box_engine.cpp` with strict C++17 flags and repeats all 122 searches. The separately written Python implementation was also rerun for $q = 61, \dots, 129$; its exact agreement with the C++ output is frozen in `finite_envelope_python_crosscheck_q061_q129.json`. The ledger checker audits that frozen cross-check, while `crosscheck_finite_backends.py` recreates it. These programs share the same exact integer-box algorithm.

For $n = 15$, the archive includes both enumerator sources, both complete outputs, the 247-profile worklist, a Burnside audit and direct witness checks. For $n = 16$, it includes the union-domain source, certificate and independent representative-list audit, as well as the two-kernel cross-check. The directory `general_sweep/` contains the general- n source, audits and

Range	Full computation	Action in the one-command verification
$q = 3, 5, 10, \dots, 129$	<code>box_engine.cpp</code>	full exact rerun and comparison with every frozen summary
$n = 15$	<code>peace15_solver.cpp</code> , <code>profile_enum.cpp</code>	profile, orbit, output-coverage and witness audits
$n = 16$	<code>peace16_union_enum.cpp</code>	domain, representative, hash, total and witness audits
$n = 8, 12, 14, 17, 18$	<code>general_sweep/solver_b.cpp</code>	worklist, orbit, output-coverage and witness audits

TABLE 3. Verification modes for the finite searches. The README gives the full rebuild commands for the computations audited only through frozen outputs here.

frozen outputs used for $n = 8, 12, 14, 17, 18$, together with the second $n = 17$ implementation and output.

The rational branch trees and support duals are standalone proof objects. The ladder summaries and exhaustive-search outputs are audit records: their logical force is the corresponding short source together with the reductions proved in Sections 6 and 7. The audits establish the necessary profile domains, the orbit and representative coverage, the candidate-space totals, the consistency and uniqueness of the output records, and the positive witnesses; they do not certify each individual completion test of the large enumerations. Among the proof-critical campaigns, $n = 15$ and $n = 17$ were decided twice by independently written enumerators, and $n = 16$ additionally by the two-kernel programs; the $n = 8, 12, 14, 18$ sweeps rest on the single audited implementation `general_sweep/solver_b.cpp`. The release command reruns the complete ladder; the larger board enumerations are reproducible with the complete build and run commands in `README.txt`, which also records the machine, compiler, per-campaign case totals and wall-clock data, and the SHA-256 digests of the frozen outputs. Emitting independently checkable UNSAT certificates for these enumerations, and a second implementation for the $n = 18$ sweep, remain natural hardening steps.

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